

Zulfiqar Ahamd Khan

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Overview: I am a postdoctoral researcher and recent PhD graduate specializing in time series data analysis, computer vision, and IoT programming. I am currently in a postdoctoral position at the Department of Computing Science, Umeå University, Sweden, funded by WASP. During my PhD, I served as Coordinator of the Intelligent Media Laboratory (IMLab), where I led a team of 8 researchers and managed multiple projects. In my current postdoctoral position, I have continued to manage research projects and co-supervise PhD students, while also contributing to publications and team coordination. My research focuses on computer vision and time-analysis. In computer vision, I develop and apply deep learning methods to address challenging perception and understanding problems in complex visual environments. I also work on active, incremental, and federated learning to develop adaptive and scalable systems, and apply Explainable AI (XAI) for model transparency and interpretability.

Summary

Personal	Professional
Date of Birth: 15 July 1995	Korean Local Patents: 06
Marital Status: Married	Total Publication: 44
Languages: English, Pashto, Urdu	First Author Publication: 14
	Total Citation: 2900+
	h-index: 22
	i ¹⁰ -index: 30+

Education

Sep 2019 - Feb 2025	Joint Master's and Ph.D. Sejong University, Seoul, South Korea Department of Software Thesis: A Study of Spatiotemporal Feature Analysis using Deep Hybrid Models for Power Forecasting
Sep 2016 - Aug 2018	M.Sc. Computer Science Islamia College Peshawar, Peshawar, Pakistan Department of Computer Science Thesis: Towards smart home automation using IoT-enabled edge-computing paradigm (DIPLab Prototype)
Aug 2014 - Aug 2016	B.Sc. Computer Science University of Malakand, Dir Lower, KPK, Pakistan Department of Computer Science

Awards & Honors

Feb 2025	Outstanding Researcher Award Sejong University, Seoul, south Korea.
Aug 2024	Global Citation Impact Recognition Recognized among the top 2% of scientists globally, as per the "Updated Science-Wide Author Databases of Standardized Citation Indicators" (2024 , 2025).
Sep 2023	Best Paper Award Next Generation Computing Conference, Best Western Premier Hotel, Jeju, South Korea
Sep 2019	100% Study Scholarship Sejong University, Seoul, South Korea, for Joint Master's and Ph.D. Degree
Aug 2019	Winner of Business Plan Competition Office of Research, Innovation and Commercialization (ORIC), Islamia College Peshawar, Peshawar, KPK, Pakistan
Dec 2019	100% Study Scholarship Islamia College Peshawar, Pakistan for M.Sc. Degree

Research Experience

May 2025 – Present	Postdoctoral Researcher (WASP) Department of Computing Science, Umeå University, Sweden - Researching and implementing open-set recognition methods - Designing and implementing Vision Transformer architectures using Spiking Neural Networks for energy-efficient and accurate visual processing.
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	- Developing federated learning pipelines for privacy-preserving computer vision on edge devices. - Developing efficient algorithms for real-time fire detection. - Guiding students in idea development, algorithm implementation, and research paper review.
March 2025 – April 2025	Postdoctoral Researcher BrainAI Lab, Kyungpook National University, Daegu, South Korea. Developed deep learning models for short-term energy forecasting and smart-grid optimization.
July 2024 – March 2025	IMLab Coordinator Sejong University, Seoul, South Korea - Leading and directing a dynamic research group of 8 members specializing in cutting-edge fields such as Energy Informatics, Computer Vision, and Signal Processing. - Providing partial supervision to Master and Ph.D. candidates, including managing Professor projects and ensuring compliance with project requirements. - Actively engaging with students, offering guidance and support in generating ideas, implementing projects, and processing research articles. - Demonstrating leadership in composing and crafting multi-year project proposals submitted to the National Research Foundation of Korea (NRF). - Facilitating collaboration and knowledge exchange within the research group to foster a creative and innovative environment. - Contributing to the advancement of the laboratory research agenda through strategic planning and effective project management.
Sep 2019 – June 2024	Research Assistant Intelligent Media Laboratory, Sejong University, Seoul, South Korea - Conducted research and development for multiple projects, fostering innovation and knowledge growth in the laboratory. - Collaborated with senior lab members to ensure smooth execution of project implementations and handled complex tasks. - Actively participated in hands-on project implementation, translating research concepts into practical outcomes.
Mar 2017– Aug 2019	Research Assistant Digital Image Processing Laboratory, Islamia College Peshawar - Oversaw research projects for bachelor students, focusing on Energy Informatics, Smart Home Solutions, Resource-Constrained Programming, and IoT. - Crafted project proposals to secure funding and resources for the laboratory's ongoing initiatives. - Actively participated in events and workshops to enhance knowledge exchange and stay updated on the latest advancements in the field.
Sep 2019– Present	Research Collaborations - Actively collaborating with research teams from diverse countries, including Saudi Arabia, Portugal, UK, Spain, Brazil, and Pakistan, fostering international partnerships. - Participating in collaborative publications and joint presentations at international Journals. - Expanding the network of research collaborations by establishing connections with institutions and researchers worldwide.

Participated Projects

Since September 2019, I have been actively involved in several key research projects funded by the National Research Foundation of Korea (NRF), where my contributions have covered a variety of critical tasks. These include projects management, implementation of advanced algorithms and methodologies, drafting and publishing research articles, and developing patents based on innovative findings. Additionally, I have been responsible for preparing comprehensive yearly reports, ensuring that project progress is accurately documented and communicated to stakeholders.

May 2025– Present	Wallenberg AI, Autonomous Systems and Software Program The Research project is funded by the Knut and Alice Wallenberg Foundation through WASP
Mar 2025– Apr 2025	Brain-Inspired AI Application (No. RS-2023-00209794) National Research Foundation of Korea (NRF) (No. RS-2025- 02218631) Institute of Information and Communications Technology Planning and Evaluation
Oct 2019– Dec 2023	Development of ensemble methods-based XAI energy platform for effective energy consumption pattern and factor analysis (2019M3F2A10731792320682075910203) National Research Foundation of Korea (NRF)
Mar 2023– Feb 2025	Anomaly behaviour recognition for accident prevention in the connected vision environment (2023R1A2C10057881220682075910102) National Research Foundation of Korea (NRF)
April 2024– Dec 2024	Development and demonstration of AI policy simulation platform technology to solve social problems in depopulated areas (RS-2024-0033958320682075910001) Institute of Information & communications Technology Planning (IITP)

Journal Publications

2026

1. Hikmat Yar, **Zulfiqar Ahmad Khan**, Adnan Hussain, Sang Il Yoon, Seoa Kim, Jungwook Choi, Chan Mi Jeon, Huisu Jeung, Kyungjung Kwon, Sung Wook Baik "[A novel deep learning framework for battery performance prediction over the operational lifespan](#)", Journal of Energy Storage
2. Adnan Hussain, Waseem Ullah, Noman Khan, **Zulfiqar Ahmad Khan**, Hikmat Yar, and Sung Wook Baik, "[Class-incremental learning network for real-time anomaly recognition in surveillance environments](#)," Pattern Recognition

2025

3. **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Hikmat Yar, Waseem Ullah, Sung Wook Baik "[Optimized Cross Module Attention Network and Database for Fire Detection](#)", Pattern Recognition
4. Hikmat Yar, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik "[EFNet-CSM: EfficientNet with a Modified Attention Mechanism for Effective Fire Detection](#)" Knowledge-Based Systems
5. Adnan Hussain, Noman Khan, **Zulfiqar Ahmad Khan**, Hikmat Yar, Min Je Kim, Sung Wook Baik, "[Edge-assisted framework for instant anomaly detection and cloud-based anomaly recognition in smart surveillance](#)" Engineering Applications of Artificial Intelligence
6. Adnan Hussain, Hikmat Yar, Noman Khan, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik "[Dual stream deep attention networks for annual population projection](#)" Pattern Analysis and Applications
7. Muhammad Munsif, Altaf Hussain, **Zulfiqar Ahmad Khan**, Min Je Kim, and Sung Wook Baik, "[Hierarchical attention-based framework for enhanced prediction and optimization of organic and inorganic material synthesis](#)," Advanced Engineering Informatics
8. Taimoor Khan, **Zulfiqar Ahmad Khan**, and Chang Choi, "[Enhancing real-time fire detection: An effective multi-attention network and a fire benchmark](#)," Neural Computing and Applications

2024

9. **Zulfiqar Ahmad Khan**, Shabbir Ahmad Khan, Tanveer Hussain, Sung Wook Baik, "[DSPM: Dual sequence prediction model for efficient energy management in micro-grid](#)", *Applied Energy*
10. Hikmat Yar, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik, "[A modified vision transformer architecture with scratch learning capabilities for effective fire detection](#)", Expert Systems with Applications
11. Adnan Hussain, Waseem Ullah, Noman Khan, **Zulfiqar Ahmad Khan**, Min Je Kim, Sung Wook Baik, "[TDS-Net: Transformer enhanced dual-stream network for video Anomaly Detection](#)", Expert Systems with Applications
12. Hikmat Yar, **Zulfiqar Ahmad Khan**, Imad Rida, Waseem Ullah, Min Je Kim, Sung Wook Baik, "[An efficient deep learning architecture for effective fire detection in smart surveillance](#)", Image and Vision Computing
13. Habib Khan, Tanveer Hussain, Samee Ullah Khan, **Zulfiqar Ahmad Khan**, Sung Wook Baik, "[Deep multi-scale pyramidal features network for supervised video summarization](#)", Expert Systems with Applications

2023

14. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Waseem Ullah, Sung Wook Baik, "[A Trapezoid Attention Mechanism for Power Generation and Consumption Forecasting](#)", IEEE Transactions on Industrial Informatics
15. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik, "[Dual stream network with attention mechanism for photovoltaic power forecasting](#)," Applied Energy
16. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Amin Ullah, Waseem Ullah, Javier Del Ser, Khan Muhammad, Muhammad Sajjad, Sung Wook Baik, "[Modelling Electricity Consumption During the COVID19 Pandemic: Datasets, Models, Results and a Research Agenda](#)," Energy and Buildings
17. Taimoor Khan, **Zulfiqar Ahmad Khan**, Chang Choi, "[Enhancing real-time fire detection: an effective multi-attention network and a fire benchmark](#)," Neural Computing and Applications
18. Hikmat Yar, Waseem Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik, "[An Effective Attention-based CNN Model for Fire Detection in Adverse Weather Conditions](#)," ISPRS Journal of Photogrammetry and Remote Sensing
19. Hikmat Yar, **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Waseem Ullah, Sung Wook Baik, "[A modified YOLOv5 architecture for efficient fire detection in smart cities](#)," Expert Systems with Applications
20. Waseem Ullah, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik, "[Sequential Attention Mechanism for Weakly Supervised Video Anomaly Detection](#)," Expert Systems with Applications

2022

21. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Fath U Min Ullah, Suneet Kumar Gupta, Mi Young Lee, Sung Wook Baik, "[Randomly initialized CNN with densely connected stacked autoencoder for efficient fire detection](#)," Engineering Applications of Artificial Intelligence
22. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Ijaz Ul Haq, Fath U Min Ullah, Sung Wook Baik, "[Towards efficient and effective renewable energy prediction via deep learning](#)," Energy Reports
23. **Zulfiqar Ahmad Khan**, Amin Ullah, Ijaz Ul Haq, Mohamed Hamdy, Gerardo Maria Mauro, Khan Muhammad, Mohammad Hijji, Sung Wook Baik, "[Efficient short-term electricity load forecasting for effective energy management](#)," Sustainable Energy Technologies and Assessments
24. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Sung Wook Baik, "[Boosting energy harvesting via deep learning-based renewable power generation prediction](#)," Journal of King Saud University-Science
25. Waseem Ullah, Tanveer Hussain, **Zulfiqar Ahmad Khan**, Umair Haroon, Sung Wook Baik, "[Intelligent dual stream CNN and echo state network for anomaly detection](#)," Knowledge-Based Systems
26. Altaf Hussain, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Fath U Min Ullah, Seungmin Rho, Sung Wook Baik, "[A hybrid deep learning-based network for photovoltaic power forecasting](#)," Complexity
27. Hikmat Yar, Tanveer Hussain, Mohit Agarwal, **Zulfiqar Ahmad Khan**, Suneet Kumar Gupta, Sung Wook Baik, "[Optimized dual fire attention network and medium-scale fire classification benchmark](#)," IEEE Transactions on Image Processing
28. Khan Muhammad, Hayat Ullah, **Zulfiqar Ahmad Khan**, Abdul Khader Jilani Saudagar, Abdullah AlTameem, Mohammed AlKhathami, Muhammad Badruddin Khan, Mozaherul Hoque Abul Hasanat, Khalid Mahmood Malik, Mohammad Hijji, Muhammad Sajjad, "[WEENet: an intelligent system for diagnosing COVID-19 and lung cancer in IoMT environments](#)," Frontiers in oncology

2021

29. Samee Ullah Khan, Ijaz Ul Haq, **Zulfiqar Ahmad Khan**, Noman Khan, Mi Young Lee, Sung Wook Baik, "[Atrous convolutions and residual GRU based architecture for matching power demand with supply](#)," MDPI Sensors
30. Hikmat Yar, Ali Shariq Imran, **Zulfiqar Ahmad Khan**, Muhammad Sajjad, Zenun Kastrati, "[Towards smart home automation using IoT-enabled edge-computing paradigm](#)," MDPI Sensors
31. Waseem Ullah, Amin Ullah, Tanveer Hussain, **Zulfiqar Ahmad Khan**, Sung Wook Baik, "[An efficient anomaly recognition framework using an attention residual LSTM in surveillance videos](#)," MDPI Sensors

2020

32. **Zulfiqar Ahmad Khan**, Amin Ullah, Waseem Ullah, Seungmin Rho, Miyoung Lee, Sung Wook Baik, "[Electrical energy prediction in residential buildings for short-term horizons using hybrid deep learning strategy](#)," *Applied Sciences*
33. **Zulfiqar Ahmad Khan**, Tanveer Hussain, Amin Ullah, Seungmin Rho, Miyoung Lee, Sung Wook Baik, "[Towards efficient electricity forecasting in residential and commercial buildings: A novel hybrid CNN with a LSTM-AE based framework](#)," *MDPI Sensors*
34. Muhammad Sajjad, **Zulfiqar Ahmad Khan**, Amin Ullah, Tanveer Hussain, Waseem Ullah, Mi Young Lee, Sung Wook Baik, "[A novel CNN-GRU-based hybrid approach for short-term residential load forecasting](#)," *IEEE Access*

Peer Review Articles

1. **Zulfiqar Ahmad Khan**, Obaidullah CZaland, Monwar Bhuyan, "FedSHIELD: Adversarial Honey-pot-Guided Client Screening Framework for Robust Federated Learning" ([IJCAI26](#))
2. **Zulfiqar Ahmad Khan**, Obaidullah Zaland, Monwar Bhuyan, "SFFormer: Saliency Aware Gated Factorized Attention for Spiking Neural Networks " ([IJCAI26](#))
3. **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Monwar Bhuyan, "Reinforcing Diversity: RL-Guided Experts for Open-Set Recognition" ([KDD26](#))
4. Md Tanvir Islam, Doheun Cha, **Zulfiqar Ahmad Khan**, Sangtae Ahn, "FRGFormer: A Firing Rate-Guided Attention Transformer for Spiking Neural Networks" (Preparing for NeurIPS)
5. **Zulfiqar Ahmad Khana**, Monowar Bhuyan, "FLARE-Net: A Federated Spiking Vision Transformer for Resource-Constrained Fire Detection" (IEEE TIP, Submitted 202)
6. Muhammad Fayaz, **Zulfiqar Ahmad Khana**, Hyeonjoon Moon , "GeoSpikeFormer: A Neuromorphic Transformer and Comprehensive Benchmark for Large-Scale Land Cover Mapping" (IEEE TIP, Submitted 202).
7. **Zulfiqar Ahmad Khan** and Sung Wook Baik "Incremental Learning Enhanced Dual-Stream Deep Attention Network for Power Forecasting" (Major Revision Submitted, Journal Applied Energy)
8. Himat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Samee Ullah Khan, Habib Khan, Min Je Kim, Sung Wook Baik "AVRNet: A Unified Deep Supervised Network for Effective Animal Voice Recognition" (Submitted to Information Processing and Management)
9. Hikmat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Je Kim, Sung Wook Baik "Attention Enhanced YOLOV8 Architecture and a Benchmark for Fire Detection in Smart Surveillance and Remote Sensing" (Submitted to Remote Sensing of Environment)
10. Waseem Ullah, Fath U Min Ullah, **Zulfiqar Ahmad Khan**, and Sung Wook Baik, Min Je Kim, Sung Wook Baik "Ensemble Feature learning with Spatiotemporal Share Attention Memory for Anomaly Detection". (Submitted to IEEE Transactions on Cybernetics)
11. Waseem Ullah, **Zulfiqar Ahmad Khan**, Hikmat Yar and Sung Wook Baik "A Dual Stream Attention Network for real world Anomalies Recognition " (Submitted to IEEE Transactions on Circuits and Systems for Video Technology)
12. Waseem Ullah, **Zulfiqar Ahmad Khan**, Hikmat Yar and Sung Wook Baik "Anomaly Recognition and Localization in Surveillance System " (Submitted to Engineering Applications of Artificial Intelligence)
13. Hikmat Yar, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Je Kim, Sung Wook Baik, "Deep Learning-Enhanced Driver Behavior Assessment with Spatial Awareness for Safe Transportation" (Submitted to Advanced Engineering Informatics)
14. Hikmat Yar, Adnan Hussain, **Zulfiqar Ahmad Khan**, Min Je Kim, and Sung Wook Baik, "Hybrid Network with Additive Attention and Explainable AI for Accurate Population Forecasting"
15. Altaf Hussain, Muhammad Munsif, **Zulfiqar Ahmad Khan**, Min je Kim, and Sung Wook Baik "A Structure-Agnostic Transformer Framework for Enhanced Materials Discovery in Engineering Informatics" (Submitted to Journal of Materials Science & Technology)

Conferences

1. Muhammad Fayaz, **Zulfiqar Ahmad Khan**, Lien Minh Dang, hyeonjoon Moon, "GeoSwin: Hierarchical Representation Learning and a Large-Scale Earth Observation Benchmark " ([CVPR26 Workshop](#) Accepted)
2. Yinuo Zhang, **Zulfiqar Ahmad Khan**, Monowar Bhuyan, "TinySEED: A Lightweight Transformer Architecture with Channel-Attention for DDoS Detection" ([SAC26](#) Accepted)
3. Obaidullah Zaland, **Zulfiqar Ahmad Khan**, and Monowar Bhuyan "Catastrophic Forgetting Resilient One-Shot Incremental Federated Learning" ([ICBDA26](#) Accepted)
4. Fath U Min Ullah, **Zulfiqar Ahmad Khan**, Sung Wook Baik, Estefania Talavera, Saeed Anwar, Khan Muhammad, "[Dual Deep Learning Network for Abnormal Action Detection](#)" 2024 IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS).
5. **Zulfiqar Ahmad Khan**, Waseem Ullah, Hikmat Yar, Noman Khan, Min Je Kim and Sung Wook Baik "[Dataset Standardization for Effective Solar Power Forecasting: A Comprehensive Analysis](#)" The 9th International Conference on Next Generation Computing (ICNGC 2023) Danang, Vietnam.
6. Noman Khan, Waseem Ullah, **Zulfiqar Ahmad Khan**, Adnan Hussain, Min Je Kim, Sang Il Yoon and Sung Wook Baik, "[Comparative Analysis of Solar Power Generation Forecasting Models for Identical Latitude Countries Data](#)". The 9th International Conference on Next Generation Computing (ICNGC 2023) Danang, Vietnam.
7. **Zulfiqar Ahmad Khan**, Noman Khan, Su Min Lee, Sang Il Yoon, Mi Young Lee and Sung Wook Baik, "[Solar Power Prediction using Dual Stream CNN-LSTM Architecture](#)" Korea Next Generation Computing Society Spring Conference 2022.
8. Habib Khan, **Zulfiqar Ahmad Khan**, Waseem Ullah, Min Jee Kim, Mi Young Lee, Sung Wook Baik, "[Efficient vehicle detection in aerial scenes](#)" Korea Next Generation Computing Society Spring Conference 2022.
9. Muhammad Munsif, Habib Khan, **Zulfiqar Ahmad Khan**, Altaf Hussain, Fath U Min Ullah, Mi Young Lee and Sung Wook Baik, "[PV-ANet: Attention-Based Network for Short-term Photovoltaic Power Forecasting](#)" The 8th International Conference on Next Generation Computing 2022.
7. Su Min Lee, Min Je Kim, Samee Ullah Khan, **Zulfiqar Ahmad Khan**, Noman Khan, Mi Young Lee and Sung Wook Baik, "[Deep Learning framework for intelligent surveillance video analytics](#)" The 7th International Conference on Next Generation Computing 2021.
8. Altaf Hussain , Samee Ullah Khan , Fath U Min Ullah , **Zulfiqar Ahmad Khan** , Mi Young Lee , Sung Wook Baik, "[Disasters Scenes Classification Based on Unmanned Aerial Vehicles Using Lightweight CNN](#)" The 7th International Conference on Next Generation Computing 2021.
9. Hikmat Yar , Tanveer Hussain , **Zulfiqar Ahmad Khan**, Lee Mi -young , Baek Sung-wook, "[Fire detection through effective vision transformers](#)" Korean Society for Next Generation Computing Journal Vol.17 No.5 2021.10 pp.21-30.
10. **Zulfiqar Ahmad Khan**, Waseem Ullah, Amin Ullah, Seungmin Rho, Mi Young Lee, Sung Wook Baik, "[An Adaptive Filtering Technique for Segmentation of Tuberculosis in Microscopic Images](#)" *NLPIR 2020: 2020 4th International Conference on Natural Language Processing and Information Retrieval*.

Patents

1. Sung-Wook Baik, RHO, Seung-Min, LEE, Mi-Young, **Zulfiqar Ahmad Khan**, Tanveer Hussain, and Amin Ullah, "Power consumption prediction system of residential and commercial building using hybrid convolution neural network and method thereof" **Registration No:** 1025005480000 (Korean Patent)
2. Sung-Wook Baik, LEE, Mi-Young, Waseem Ullah, Tanveer Hussain, and **Zulfiqar Ahmad Khan**, "Anomaly recognition method and system based on LSTM" **Registration No:** 1026012330000
3. Sung-Wook Baik, LEE, Mi-Young, **Zulfiqar Ahmad Khan**, Tanveer Hussain, Kim Minje, "Method for predicting renewable energy consumption and production based on deep learning and apparatus thereof" **Application No:** 1020220061428 (Submitted)

4.

Sung-Wook Baik, LEE, Mi-Young, Altaf Hussain, **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Kim Minje, and Lee Sumin “Method for predicting renewable energy consumption and production based on deep learning and apparatus thereof” **Registration No:** 1025335080000

5.

Sung-Wook Baik, LEE, Mi-Young, **Zulfiqar Ahmad Khan**, Kim Minje, Yoon Sangil, and Lee Sumin “Method for predicting energy production and consumption data using double sequence deep learning model, and apparatus thereof” **Registration No:** 1025218070000

6.

Sung-Wook Baik, LEE, Mi-Young, **Zulfiqar Ahmad Khan**, Fath U Min Ullah, Kim Minje, Yoon Sangil, and Lee Sumin “method for predicting energy consumption and production based on using hybrid convolutional neural network and regression neural network and apparatus thereof” **Registration No:** 1025218080000

Technical Skills

- Programming Languages: Python, C++, PhP, etc.
- Deep Learning Frameworks: TensorFlow, Keras, PyTorch, etc.
- Computer Vision Tools: OpenCV, Scikit-image, Pillow, etc.
- Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, etc.
- Database Management: XAMP, MySQL, SQLite
- IoT Technologies: MQTT, Raspberrry Pi, Arduino
- Version Control: Git, GitHub
- Development Environments: Spyder, Geany, Jupyter Notebook, etc.
- Machine Learning Algorithms: Supervised and unsupervised learning, active learning, incremental learning, etc.
- Statistical Analysis: Time series data analysis, regression models, data refinement methods, etc.
- Simulation Software: MATLAB Simulink
- Document Processing: LaTeX, Microsoft Office

Member of Reviewer Board in Multiple Publisher (60+ reviews)

- IEEE Transactions on Cybernetics
- IEEE Transactions on Industrial Informatics
- Journal of King Saud University - Computer and Information Sciences
- Engineering Application of Artificial Intelligence
- Knowledge Based System.
- Information Processing and Management
- Artificial Intelligence Review
- Cluster Computing
- Energy Efficiency
- Stochastic Environmental Research and Risk Assessment
- Scientific Report
- IET Generation, Transmission & Distribution
- IET Renewable Energy
- detailed review list ([ORCID](#))

References

Prof. Sung Wook Baik | Ph.D. Supervisor

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Dr. Amin Ullah | LAB Coordinator

- Senior AI/ML Software Engineer
- Boeing Research & Technology Software, Seattle, WA, USA
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